

Shiny Overview

Nanocourse Lecture 1, 8/4/25

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TAs: Sriya & Ermis!

Nanocourse Intro

Goal

Everyone gets a cool app built in the next 2 days!

Scope

Introduce core Shiny concepts quickly & catalyze further independent learning.



Nanocourse schedule

- Day 1
 - Overview
 - UI side
 - Server side
 - Case study
 - Project time
- Day 2
 - Interactive plots & extras
 - Deployment
 - Project time
 - Project presentations!



Shiny Showcase

1. [Kmeans](#)
2. [Classic dashboard](#)
3. [Fancy external example](#)
4. [TO design](#)

[Shiny Gallery](#)



What are Shiny use cases?

“In short, Shiny gives you the ability to pass on some of your R superpowers to anyone who can use the web.”

– Mastering Shiny

- Exploring complex data
- Sharing complex data
- Teaching quantitative concepts
- Making “tools” for the lab



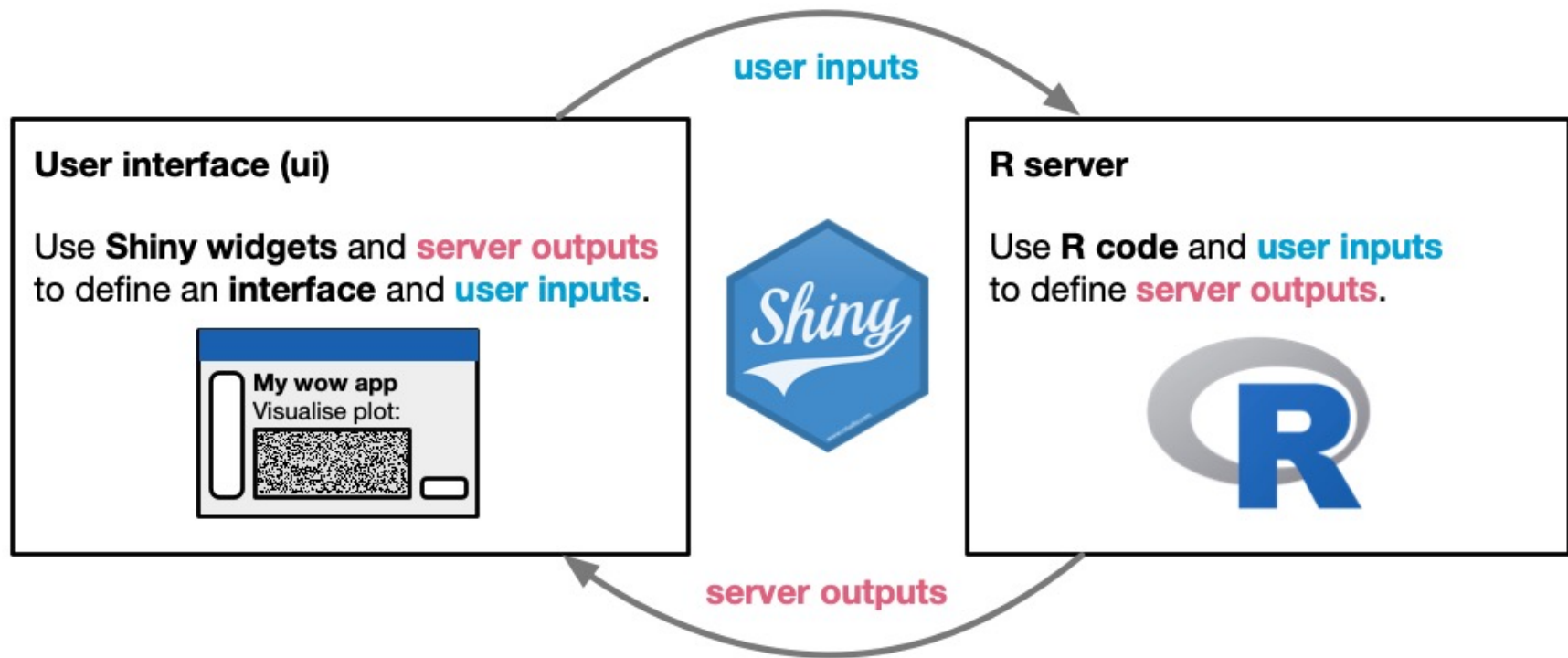
5 min brainstorm



Without any code, what
do you want your project
app to do?

Draw on the whiteboard!

Shiny = server + interface



Warning: Shiny coding may seem weird...

Programming Styles	
Imperative	• What we use in data analysis • Load data, wrangle, visualize, model, save • Functions written – do this now • “Make me a sandwich” • Specific commands executed immediately • Code executed in top to bottom order
Declarative	• What we use in Shiny apps • Express higher-level goals • Functions written – done when needed • “Ensure that there is a sandwich in the refrigerator whenever I look inside” • Code executed when new input available

```
#functions
```

```
a()
```

```
b()
```

```
c(arg_1 = 1)
```

```
a(
```

```
  b(
```

```
    c(arg_1 = 3)
```

```
  )
```

```
)
```

https://bookdown.org/pdr_higgins/rmrwr/a-gentle-introduction-to-shiny.html

Hello World

Rstudio > File > New File > Shiny web app

```
# UI

plotOutput(outputId = "distPlot")

sliderInput("bins", min = 1, max = 50)
```

```
# Server

output$distPlot <- renderPlot({
  bins <- seq(min, max, input$bins)
  hist(..., breaks = bins)
})

# Run app

shinyApp(ui = ui, server = server)
```



Shiny inputs

Inputs

Inputs allow users to interact with the webpage by clicking a button, entering text, selecting an option, and more.

The inputs shown here are just a sample of the many inputs available in Shiny. For more, see [awesome Shiny extensions](#).

The image displays a collection of 12 Shiny input components arranged in a 4x3 grid. Each component is shown with a title and a small icon. The components are: File Input (with a 'Browse...' button and 'No file selected' text), Numeric Input (with a text box containing '100'), Password Field (with a text box containing 'Enter password' and a password icon), Plot (showing a scatter plot of data points), Radio Buttons (with two options: 'Give you up' and 'Let you down'), Select (Multiple) (with a text box), Select (Single) (with a dropdown menu showing 'Choice 1A'), Selectize (Multiple) (with a text box), Selectize (Single) (with a dropdown menu showing 'Choice 1A'), Slider (with a range from 0 to 20 and a slider handle at 10), Slider Range (with a range from 1 to 100 and two slider handles at 25 and 70), and Submit Button (with a blue 'Submit' button).

Can you change the old faithful slider to a numeric input?

```
# UI
```

```
sliderInput()  
numericInput()
```

<https://shiny.posit.co/r/components/>

Shiny outputs

Outputs

Outputs create a spot on the webpage to display results from the server, such as text, tables, plots, and more.

The outputs shown here are a small sample of Shiny outputs available in R. For more, see [htmlwidgets](https://htmlwidgets.github.io/).

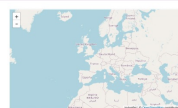
DataTable

This	That
And	The
Other	Thing

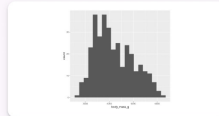
Image



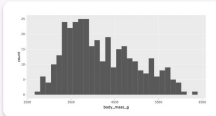
Map (leaflet)



Plot (ggplot2)



Plot (plotly)



Table

species	island	bill_length_mm	bill_depth_mm
Adelie	Torgersen	39.10	18.70
Adelie	Torgersen	39.50	17.40
Adelie	Torgersen	40.30	18.00
Adelie	Torgersen	NA	NA
Adelie	Torgersen	36.70	19.30

Table (gt)

Penguins in the Palmer Archipelago					
sex	bill_length_mm	bill_depth_mm	flipper_length_mm	body_mass_g	
Biscoe					
Adelie	female	37.75	17.70	181.0	3375.0
Adelie	male	49.00	18.90	197.0	4050.0
Gentoo	female	45.50	14.25	212.0	4700.0
Gentoo	male	49.00	15.70	221.0	5000.0

Table (reactable)

species	island	bill_length_mm	bill_depth_mm
Adelie	Torgersen	39.1	18.7
Adelie	Torgersen	39.5	17.4
Adelie	Torgersen	40.3	18
Adelie	Torgersen		
Adelie	Torgersen	36.7	19.3

Text

UI

☒ Show slider

1 2 3 4 5 6 7 8 9 10

Value Box



Verbatim Text

Can you replace the plot with Verbatim Text output of the faithful dataset?

```
# UI
```

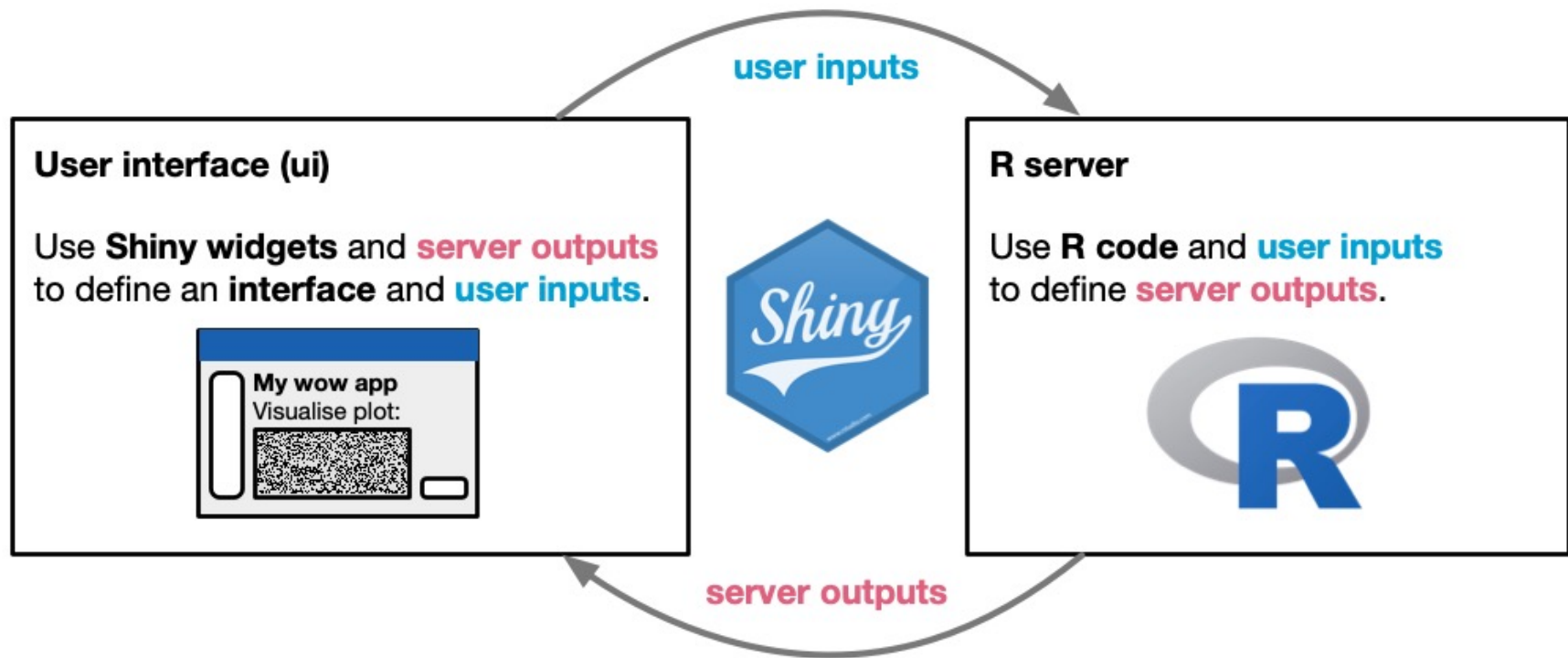
```
plotOutput()  
verbatimTextOutput()
```

```
# Server
```

```
renderPlot({})  
renderPrint(faithful)
```

<https://shiny.posit.co/r/components/>

Shiny = server + interface



Quiz 1/3: Display different parts of dataframe!

Input

Select (Single)

Choice 1A

Output

Table

species	island	bill_length_mm	bill_depth_mm
Adelie	Torgersen	39.10	18.70
Adelie	Torgersen	39.50	17.40
Adelie	Torgersen	40.30	18.00
Adelie	Torgersen	NA	NA
Adelie	Torgersen	36.70	19.30

```
# UI
```

```
selectInput(inputId = "select_1",  
             choices = c("A", "B", "C") ),
```

```
tableOutput(outputId = "table_1")
```

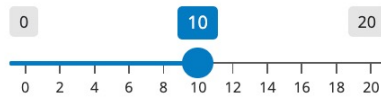
```
# Server
```

```
output$table_1 <- renderTable({  
  df %>% filter(col == input$select_1) })
```

Quiz 2/3: Move a line on a plot!

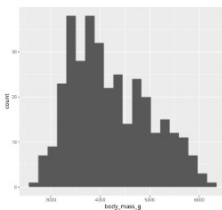
Input

Slider



Output

Plot (ggplot2)



```
# UI
```

```
sliderInput(inputId = "slide_1",  
            min = 1, max = 1000, value = 10),
```

```
plotOutput(outputId = "plot_1")
```

```
# Server
```

```
output$plot_1 <- renderPlot({  
  ggplot() +  
    geom_hline(yintercept = input$slide_1) +  
    scale_y_log10(limits = c(1,1000)) })
```

Quiz 3/3: Recreate the most used bioinfo tool!

Input

Text Area →

Output

Text →

Reverse Complement

Reverse Complement converts a DNA sequence into its reverse, complement, or reverse-complement counterpart. You may want to work with the reverse-complement of a sequence if it contains an ORF on the reverse strand.

Paste the raw or FASTA sequence into the text area below.

- Convert the DNA sequence into its reverse-complement counterpart.

[\[home\]](#)

Quiz 3/3: Recreate the most used bioinfo tool!

Input

Text Area



Enter text

Output

Text



Enter text

```
Library(Biostrings)

# UI

textAreaInput(inputId = "seq_1",
               value = "AATTCCGG"),

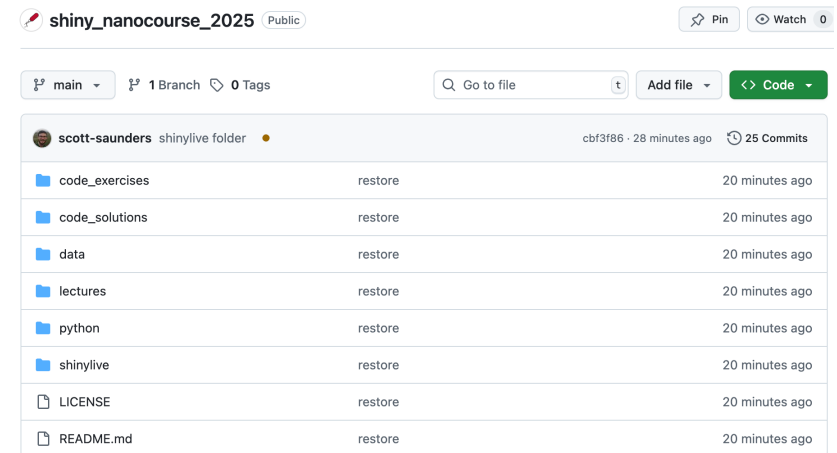
textOutput(outputId = "rev_comp_1")

# Server

output$plot_1 <- renderText({
  seq_complement(seq_reverse(dna(input$seq_1)))
})
```


Code examples: Hello github!

- Go to course github
- Download `code_exercises/L1_..._app.R` (or clone whole repo)
- Test app functionality
- Follow commented instructions in code
- Check solution in `code_solutions/`



github.com/saunders-lab/shiny_nanocourse_2026

Day 1

- Shiny Overview (Scott)
- UI side (Sriya)
- Server side (Ermis)
- Lunch
- Case study (Scott)
- Projects



Reactivity motivation

